

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1370

Goldmine 100 kV
December 30, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1370. The Interconnection Customer proposes a 343.4 MW solar photovoltaic facility interconnecting at the Goldmine 100 kV substation in the Illinois area. The request was received on December 29, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 518 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,014,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GLENN / GOLDMINE 100kV	PEACHTON / GOLDMINE 100kV Line	251.3	117	100.0
GOLDMINE / PEACHTON 100kV	GLENN / GOLDMINE 100kV	216.7	135	100.0
GLENN / CHURCH ROAD 100kV #1	GLENN / CHURCH ROAD 100kV line	114.1	141	26.4
GLENN / CHURCH ROAD 100kV Ckt 2	GLENN / CHURCH ROAD 100kV Ckt 1	114.1	141	26.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GOLDMINE / GLENN 100kV	Rebuild line with higher-capacity conductor	\$415,000	100% (\$415,000)
GOLDMINE / PEACHTON 100kV	Rebuild line with higher-capacity conductor	\$1,919,000	100% (\$1,919,000)
GLENN / CHURCH ROAD 100kV #1	Rebuild line with higher-capacity conductor	\$2,044,000	77% (\$1,566,000)
GLENN / CHURCH RD 100kV Line 2	Rebuild line with higher-capacity conductor	\$1,455,000	77% (\$1,114,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$5,014,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.